ASAD HUSSAIN

Software Engineering | Applied AI and Machine Learning | Backend and Web Development

Islamabad, Pakistan | +92 320 4141092 | asadh1521@gmail.com

LinkedIn | GitHub | Software portfolio

PROFESSIONAL SUMMARY

Software and AI developer building full-stack web applications, Python APIs and automated data workflows. Automation Engineer at Cymatics Technologies, where I completed Bid Monitor. Hands-on work spans relational databases, authentication, LLM integration, machine learning, automated testing and cloud deployment.

TECHNICAL EXPERTISE

Languages: Python, JavaScript, PHP, Java, SQL

Web and backend: FastAPI, Flask, Laravel, Spring Boot, Vue.js, HTML/CSS, REST APIs

Data and cloud: PostgreSQL, MySQL, Supabase, SQLite, Docker, GitHub Actions, Render, Vercel

AI and machine learning: Scikit-learn, TensorFlow/Keras, PyTorch, XGBoost, OpenCV, YOLOv8

Agents and automation: Gemini API, RAG, LangChain, LangGraph, CrewAI, n8n, Playwright

Engineering practices: Git, pytest, PHPUnit, API integration, debugging, functional testing

SOFTWARE AND AI EXPERIENCE

Automation Engineer — Sep 2026 - Present

CYMATICS TECHNOLOGIES | Full-time | Remote

- Delivered Bid Monitor: Python/FastAPI collectors, HTML/PDF extraction, deduplication and a searchable web dashboard backed by Google Sheets.
- Integrated procurement sources, bid amounts and evidenced contact records; scheduled collection with GitHub Actions and connected reviewed n8n outreach workflows.

Backend AI Engineering Intern — Jul - Sep 2026

FlyRank AI | Chicago, USA | Remote

- Built authenticated REST CRUD APIs with relational persistence and Docker; developed a PostgreSQL-backed usage-metering and billing capstone.
- Implemented retry-safe usage tracking, quota enforcement and signature-verified Stripe test-mode webhooks; documented 27 passing automated tests.

Full Stack Development Intern — Jul - Aug 2026

Lantrotech | Islamabad

- Developed PHP/Laravel web applications with SQL persistence, backend integration and procurement-data automation.

Machine Learning Intern — Jun - Jul 2026

CodeAlpha | Remote

- Developed credit-scoring, character-recognition and disease-prediction models; evaluated classification with precision, recall, F1 and ROC-AUC.
- Built RAVDESS speech-emotion recognition using MFCC features and MLP/CNN models.

AI Model Evaluator — Nov 2025 - Jan 2026

JSM Consulting INC | Sunnyvale, USA | Remote

- Reviewed AI-generated pull requests and coding-agent sessions; documented defects and technical rationale, and built Docker-based evaluation benchmarks.

SELECTED SOFTWARE AND AI PROJECTS

MarketHub | Multi-vendor E-commerce

Laravel 12, Vue 3, Inertia.js, MySQL, Docker | GitHub

- Built customer, vendor and admin workflows with role-based access, product search, inventory, carts, orders and transactional demo checkout.
- Integrated a managed MySQL database, REST catalogue API and PHPUnit feature tests; deployed on Render. Payments are simulated.

Career Atlas | AI Career Assistant

FastAPI, Flutter, PostgreSQL, Gemini, LangGraph, Playwright | GitHub

- Built job-feed ingestion, PDF CV extraction, AI matching and tailored application drafts with deduplication and a persistent review dashboard.
- Added browser-form adapters, bounded retries and submission evidence, with separate tracking for drafts, attempts and confirmed submissions.

AutoFlow | Business Process Automation

Java Spring Boot, Python FastAPI, PostgreSQL, n8n, JavaScript | GitHub

- Connected a responsive dashboard to customer CRUD, webhook-triggered workflows, execution tracking, CSV cleaning and reporting services.
- Packaged Java and Python services with Docker, GitHub Actions CI and a Render deployment blueprint.

CanopyNet | Forest Cover Classification

TensorFlow, Keras, Scikit-learn, XGBoost

- Compared a tuned MLP, Random Forest and XGBoost on 581,012 samples with 54 features and seven classes; achieved 95.93% test accuracy and 0.94 macro F1.

AI Support Triage | Retrieval and Tool Use

FastAPI, Hugging Face, RAG, Firecrawl, n8n, Docker | GitHub

- Built a support-agent prototype combining policy retrieval, order lookup, source references and decision traces, with human escalation and failure-handling tests.

ProjectPulse | Collaborative Task Management

Laravel, Blade, JavaScript, Sanctum, SQL, PHPUnit | GitHub

- Implemented a drag-and-drop Kanban board, team permissions and token-secured REST API; integrated cached holiday data, validation, pagination and automated tests.

ADDITIONAL SOFTWARE EXPERIENCE

Software Engineer Jun - Sep 2025

Nelson Bradx LLC | Remote

- Developed web application features using Rails, Spring Boot, Vue.js and ExtJS; supported deployment with Docker and Linux.

Application Development Intern Jun - Aug 2025

Codes Apex | Islamabad

- Built PHP/Laravel features and supported data integration, debugging, automation and functional testing.

EDUCATION AND PROFESSIONAL DEVELOPMENT

National University of Sciences and Technology (NUST)

B.E. Electrical Engineering | 2023 - 2027 (expected)

atomcamp AI Bootcamp: deep learning, NLP, generative AI, RAG, LangChain, n8n and MLOps.

AI Apps on Azure - Microsoft/edX (2025) | Data Analysis with Python - IBM (2024)